

RMSD Workflow Progress Summary

Date: 2026-07-22

1) Scope Completed

- Refined and simplified both RMSD tools for beginner-friendly command usage.
- Improved command conventions to reduce ambiguity around reference and target frame roles.
- Standardized documentation language for matrix orientation and mode behavior.

2) Script Updates

- RMSD_per_frame_biopython_7-21-26.py: annotations expanded with clearer scenarios, fixed command ordering guidance, and matrix-vs-list mode distinctions.
- RMSD_calculation_function_7-21-26.py: header annotations expanded with purpose, scenarios, argument explanations, and RMSD mathematical formula.
- Added explicit backbone-only explanations and practical usage notes in both script docs.

3) README Improvements

- README.md upgraded into a mini cookbook with detailed scenario walkthroughs.
- Added explicit output expectations (default filenames, list vs matrix output types).
- Added formula section: $\text{RMSD} = \sqrt{(1/N) * \sum(|r_i - s_i|^2)}$.
- Added troubleshooting for common Windows and dependency issues.
- Clarified singular vs plural reference flags:
 - --reference-frame => list output (Frame,RMSD_Angstrom)
 - --reference-frames => matrix output (references as rows, targets as columns)

4) Orientation and Mode Rules (Now Explicit)

- To force Excel-style orientation (vertical references, horizontal targets), use matrix mode:
 - --reference-frames ... --targets ...
- Using --reference-frame ... --targets ... does not create a matrix; it creates a simple list CSV.
- This was identified as a key confusion point and corrected in docs.

5) Commands and Execution Support Delivered

- Provided multiple copy-pasteable PowerShell commands for 3vs1, 1vsN, NvsM, all-vs-all, and atom-filtered scenarios.
- Created output folder: test_outputs_for_RMSD calculations.
- Explained how to include self-comparisons (e.g., frame 10 vs frame 10) using matrix mode.

6) Environment and Error Resolution

- Diagnosed python PATH alias issue on Windows; switched to py launcher usage.
- Verified Python launcher and environment (py -3.13).
- Resolved missing Biopython dependency and confirmed installation (Biopython 1.87).
- Explained harmless SyntaxWarning from backslashes in docstring examples.

7) Validation Status

- Syntax/error checks run on edited Python files after modifications.
- No code errors reported for the updated RMSD scripts after final edits.

8) Current Outcome

- The workflow now supports beginner-safe command patterns with consistent reference/target semantics.
- Documentation clearly separates list mode vs matrix mode and prevents axis-orientation confusion.
- User can now generate reproducible RMSD outputs for classroom and research reporting with minimal setup.